

Arnav Panigrahi

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Experience

- Software Developer**, AllCheer – California, USA Mar 2025 – Present
- Built a HIPAA compliant LangChain agent-driven ROI automation orchestrating stakeholder authorization, compliance validation, and document distribution, enforcing multi-party approval logic and reducing manual processing time by 60%.
 - Engineered a containerized FastAPI microservice that emulates authenticated user sessions against RethinkBH endpoints to fetch records directly, bypassing public APIs with 24–48 hour delays and enabling near real-time retrieval.
 - Built a React + TypeScript internal analytics dashboard integrating OpenAI APIs and NLP pipelines to extract structured entities from unstructured therapist notes, improving clinical scheduling insights and achieving 92% extraction accuracy.
- Software Developer Intern**, AllCheer – California, USA Jun 2024 – Dec 2024
- Developed a Python-based logistics system integrating Google Maps Distance Matrix and Routing APIs to compute multi-stop schedules for a 20-provider workforce (1500 routes/month), exposing route computation via REST APIs.
 - Designed REST API schemas and backend integration layers enabling the logistics engine to interact with internal databases and scheduling systems while validating workflow automation prototypes through rapid R&D using Make.com.
- Junior Software Developer**, Pinnacle Consulting LLC – Bhubaneswar, India Jan 2022 – Aug 2023
- Integrated ArcGIS Online geospatial services into Angular and .NET enterprise dashboards for a \$75M utility provider, enabling real-time visualization of infrastructure assets and reducing operational oversight errors by 30%.
 - Delivered 3 ERP platforms using Node.js, Angular, and SQL Server, implementing JWT-based RBAC, optimized relational queries, and modular service architecture to automate invoicing and reporting workflows, improving efficiency by 25%.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C#, Java
Backend: FastAPI, Node.js (Express), REST APIs, WebSockets, Microservices, Authentication (JWT/OAuth)
Frontend: React, SvelteKit, Angular, TypeScript, TailwindCSS, Vite
Cloud & DevOps: Google Cloud, AWS, Docker, CI/CD pipelines, Serverless architectures
Databases: PostgreSQL, SQL Server, MongoDB, Redis, Pinecone (Vector DB)
AI Systems: LangChain Agents, Retrieval-Augmented Generation (RAG), LLMs

Projects

- LangChain RAG System for Knowledge Retrieval** | LangChain, OpenAI, Qdrant, FastAPI 2026
- Engineered a production-ready RAG system with config-driven ingestion, semantic chunking, OpenAI embeddings, and Qdrant vector search (top-k optimization), enabling low-latency, context-aware QA over domain-specific corpora.
 - AI assistant-style retrieval workflows with LangChain, hardening against injection, source attribution, retry logic, and lazy initialization, and validated system reliability through mocked unit and real-API integration testing.
- Nier Archive (Real-Time Site)** | SvelteKit 5, TypeScript, Cloudflare Workers, Durable Objects 2026
- Engineered a globally distributed web application using SvelteKit and Cloudflare Workers, implementing stateful WebSocket sessions via Durable Objects to enable real-time cursor presence across clients.
 - Streamlined content delivery by building a custom Vite plugin for build-time markdown parsing and deploying the application globally using Cloudflare Pages and KV storage for edge-optimized performance.
- bedtime.ai – Multi-Modal AI Storytelling Platform** | Phi-3, EfficientNet, TTS, FastAPI, React 2025
- Built a multi-modal AI assistant that transforms user sketches into narrated stories using fine-tuned EfficientNet classification, Phi-3 LLM generation, and Coqui TTS within a real-time inference pipeline, deployed on Google Cloud
 - Developed an end-to-end ML system with dataset preprocessing, model fine-tuning (250 classes), and prompt engineering for Phi-3, orchestrated via FastAPI with optimizations for inference latency, accuracy, and user experience

Education

University of California, Riverside Sep 2023 – Mar 2025
MS in Computer Science
Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Computer Architecture, Operating Systems